Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

A. 3.0 amps, 4 ohms

B. 2.8 amps, 3.2 ohms

C. 3.75 amps, 3.2 ohms

D. 4 amps, 3 ohms

E. 3.33 amps, 3.6 ohms

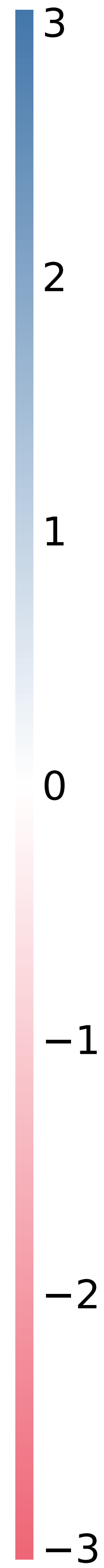
F. 5 amps, 2.4 ohms

G. 4.5 amps, 2.67 ohms

H. 2.2 amps, 5.45 ohms

I. 2.5 amps, 4.8 ohms

J. 3.5 amps, 4 ohms



Reasoning + Answer:

<think> We are given that the automobile battery produces a potential difference of 12 volts and a headlight bulb dissipates 40 watts of joule heat when connected directly across the terminals. First, we need to find the current drawn by the bulb using the formula P = V^2/R. However, we don't know the resistance, so we will first rearrange the formula to R = V^2/P, which gives us R = 12^2 / 40. We can now solve this equation. Next, plugging this value of resistance into Ohm's law I = V/R, we can calculate the current drawn by the bulb. Once we have these values, we can check the options that match our results. Given P = 40, we calculate R = 144 / 40 = 3.6 and I = V/R = 12 / 3.6 = 3.33. So we have that current of 3.33 amps and a resistance of 3.6 ohms. Since the potential difference (voltage) here was given and current and resistance are calculated, answer E = 3.33 amps, 3.6 ohms </think><answer> 3.33 amps, 3.6 ohms </answer>